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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

AMCS202S – ALLIED MATHEMATICS – II

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Find $L\left(e^t + \frac{1}{et}\right)^2$.
2. Find the value of L[*3cos 4t*].
3. Find the vector nuemal to the surface $x^2 + y^2 + 2z^2 = 4$ at the point (1,1,1).
4. State Green's theorem in the plane.
5. Define Interpolation.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find the Laplace Transforms of
   i) $\cos^2 t$ ii) $\cos^2 2t$
7. Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = e^{-x}$, using Laplace Transform, given that y(0)=0, y'(0) = 1.
8. Find the directional derivative of the function $xy + yz + zx$ at (1,1,3) in the direction $i + 2j + 2\bar{k}$.
   .
9. Evaluate $\iint_S x\,dydz + ydzdx + z\,dxdy$ over the surface of the sphere $x^2+y^2+z^2 = a^2$.
10. Find u_6, given that

u_0	u_1	u_2	u_3	u_4	u_5
25	25	22	18	15	15

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Evaluate (i) $L[e^{-3t} \sin t \cos t]$ (ii) $L[e^{-5t} \cos^2 t]$.

(or)

b) Evaluate $L^{-1}\left[\frac{s^2-1}{(s^2+1)^2}\right]$.

12. a) Solve $\frac{d^2y}{dx^2}$ +4y = $5e^{-x}$, using Laplace Transform, given that y(0) = 2, y'(0) = 3.

(or)

b) Using Laplace Transform, Solve $\left(D^2+2D-3\right)y=\sin t$ given that $y=\frac{dy}{dt}=0$ when $t=0$.

13. a) Find curl $\overline{F}$ in the following cases:
i) $\overline{F}$ = $x^2y\overline{i}$ + $y^2z\overline{j}$ + $z^2x\overline{k}$
ii) $\overline{F}$ = xyz$\overline{i}$ + $xyz^2\overline{j}$ + $x^2yz\overline{k}$

(or)

b) Show that $\nabla^2 \log r = \frac{1}{r^2}$.

14. a) Verify Green's theorem for $\int_C (3x^2 - 8y^2)dx + (4y-6xy)\,dy$, where C is the boundary of the region enclosed by the parabolas x^2 = y and y^2 = x.

(or)

b) Using Stoke's theorem, evaluate $\int_C (\sin z\,dx - \cos x\,dy + \sin y\,dz)$, where C is the boundary of the rectangle $0 \le x \le \Pi$, $0 \le y \le 1$, z = 3.

15. a) Use Newton's formula to find y when x = 142, given that

x	140	150	160	170	180
y	3.685	4.854	6.302	8.076	10.225

(or)

b) Use Lagrange's formula to find y when x = 2, given that

x	0	3	5	6	8
y	276	460	414	343	110

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